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Applying the critical speed concept to racing strategy and interval training prescription

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ABSTRACT

The use of personal records (PRs) for running different distances may be used to derive critical speed (CS) and the finite capacity for running speeds exceeding CS (D'). Using CS and D' , individualized speed-time and distance-time relationships can be modeled (i.e., time limits associated with running at a given speed or a given distance can be derived via linear regression with a high degree of accuracy). The running 3-min all-out exercise test (3 MT) has emerged as a method for estimating CS and D' on a large group of athletes in a single visit. Such a procedure is useful when PRs are not readily available (e.g., team sport athletes). This article reviews how to administer and interpret the running 3 MT, how CS and D' can inform racing strategy, and how CS and D' can be used to prescribe and evaluate high-intensity interval training (HIIT). Directions for deriving HIIT bouts using either fixed-distances or fixed-speeds are provided along with CS dose-responses to short-term HIIT programs.

KEYWORDS: 3-min all-out exercise test, Anaerobic power, Critical power, High-Intensity, Severe Domain

Introduction

Team sports such as rugby, lacrosse, field hockey, and football utilize both anaerobic and aerobic systems intermittently, and often to their maximal limits.¹ Therefore, high-intensity interval training (HIIT) programs are suitable for conditioning these athletes. The critical speed (CS) concept has emerged as a method to quantify how the aerobic and anaerobic energy systems coalesce to support high-intensity performance and HIIT.² Emergence of the 3-min all-out exercise test (3 MT) has amplified research on the topic of CS in running and on the critical power concept in cycle ergometry.³

The severe exercise domain is characterized by an intensity too severe to permit a steady-state of oxygen uptake (VO_2), resulting in the reliance of anaerobic energy pathways.³ The basis for prescribing HIIT in the severe exercise domain begins by determining an athlete's CS. Critical speed is a mechanical measure that has been suggested to represent the maximal lactate or VO_2 steady state.⁴ Metabolic steady-states of VO_2 are unattainable $>\text{CS}$ and may evoke values approaching and achieving maximum VO_2 ($\text{VO}_{2\text{max}}$).³ The pronounced time-dependent rise in VO_2 toward $\text{VO}_{2\text{max}}$ during constant-speed running is termed the VO_2 slow component, an event largely attributed to the recruitment of type II muscle fibers.⁵ Indeed, by engaging in HIIT in the severe domain, improvements in CS and overall running performance can be achieved,⁶ resulting in attenuation of the VO_2 slow component and less reliance of type II muscle fibers.⁷

This review describes procedures for modeling the speed-time and distance-time relationships from prior personal record (PR) running performances or a running 3 MT. Procedural considerations of the running 3 MT are also discussed. The modeling of performance data from two former Olympic runners are used to explain the utility of CS concept for

developing racing strategy. Finally, a simple application of the CS concept on prescribing and assessing HIIT is summarized.

Modeling Maximal Performance Capacity

The CS concept originated as a method of modeling the total work capacity of an isolated exercising muscular group relative to the time of force decline.⁸ The method was adapted subsequently for leg cycle ergometry⁹ and slightly later for running.¹⁰ In running, it is common to derive CS and the finite capacity for running speeds exceeding CS (D') from PR times of different distances,¹⁰ although exhaustive times for runs of different supramaximal running speeds on a treadmill have been used.¹¹ Take for example, Steve Prefontaine (Pre), a middle distance runner who during the late 1960's and early 70's, held every USA distance running record between 2,000 m and 10,000 m. Plotting Pre's PR times over various race distances indicates his CS was $6.0 \text{ m}\cdot\text{s}^{-1}$ and his D' was 200 m (Figure 1, panel A). Figure 1, panel A reflects Pre's PRs associated with four running distances (D). Using linear regression, the distance associated with a given t_{LIM} may be resolved using:

$$D = (CS \cdot t_{\text{LIM}}) + D', \quad (\text{Eqn 1})$$

where D and D' are in m, CS is in $\text{m}\cdot\text{s}^{-1}$, and t_{LIM} is in s. The t_{LIM} for a given distance may be resolved using:

$$t_{\text{LIM}} = (D - D')/CS. \quad (\text{Eqn 2})$$

A desired running speed (S) for a given t_{LIM} may be resolved using:

$$S = D'/t_{\text{LIM}} + CS. \quad (\text{Eqn 3})$$

Modeling high-intensity performance for an individual athlete also may be projected from a single 3 MT. The 3 MT is particularly useful for team sport athletes, where PR times of different distances are not readily known. Moreover, the modeled t_{LIM} values can be used to prescribe and monitor responses to HIIT. The utility of the 3 MT is that the parameters of CS and D' may be derived on a group of athletes in a short period of time.¹²

Summary of the Running 3-min all-out Exercise Test

The 3 MT emerged as a method of deriving estimates of critical power and the work capacity exceeding critical power (W') in cycle ergometry.^{13, 14} The 3 MT procedures developed subsequently for running entail having the subject build up their running speed to avoid muscle strain (e.g., hamstrings) until reaching their maximal speed.¹² Once maximal speed is achieved, the athlete attempts to sustain that speed for as long as they can, and is instructed to run all-out until told to stop, typically at a time of 3:05 to ensure a full sampling of 3 min of data. Our subjects are counseled to expect that about half way through the 3 MT, they will experience a sense of demoralization. Despite the urge to quit, subjects are encouraged to keep going. The objective of the test is to wholly deplete and measure the finite running capacity above CS. Such a measurement cannot be achieved if a subject “holds back” their best effort.

Speed or distance relative to time is obtained through various manners of recording [e.g., global positioning sensor (GPS) technology,¹² accelerometers,¹¹ video recording⁶], and these data are subsequently evaluated using Eqn 4 (below). To help illustrate the mathematics of the 3 MT, Pre’s data from Figure 1 is used as an example. His initial speed at 150 s of the 3 MT (S_{150s}) would have been approximately $7.33 \text{ m}\cdot\text{s}^{-1}$; whereas his average speed between 150 and 180 s (i.e., CS) would have been $6.0 \text{ m}\cdot\text{s}^{-1}$ (Figure 1). The solution of his D' would be derived using:

$$D' = 150 \text{ s } (S_{150s} - CS), \quad (\text{Eqn 4})$$

where Pre's D' would have been $200 \text{ m} = [(7.33 \text{ m}\cdot\text{s}^{-1} - 6.0 \text{ m}\cdot\text{s}^{-1})] \cdot 150 \text{ s}$. Interestingly, the solution of CS and D' from the 3 MT also can be derived by calculating the slope and y-intercept of the distances at 150 s and 180 s, respectively, as shown conceptually in Figure 1, panel A.

Pacing is of concern in 3 MTs involving procedures that allow the subject to control their intensity or resistance relative to time (e.g., pace selection in running, gear selection in cycling on the CompuTrainer¹⁵). Conversely, fixed-loads for cycle ergometry are derived using a linear factor from a preliminary incremental exercise test^{13, 14} or a fixed percentage of body mass.¹⁶⁻¹⁸ For such 3 MTs, the load is “inescapable,” similar to the Wingate Anaerobic Power Test,¹⁹ whereby the subject pedals against a fixed load for a fixed amount of time. With the running 3 MT, athletes have the option to hold back and pace to ensure a stronger finish, although they are instructed not to do so. As the CS is estimated using the end speed of the test, if an athlete paces and a sudden end-spurt in speed ensues, that athlete will not truly expend all of D' during the initial 150 s, resulting in an inflated estimate of CS and a low estimate of D' . Having evaluated the running 3 MT on over 400 subjects, I have tested and re-tested numerous subjects suspected of excessive pacing. Of those re-tested, the paced test is characterized by a nadir of speed at their true CS between ~120 and 160 s, with the end spurt occurring between 160 and 180 s. When such a speed-time profile is observed, re-testing when possible is recommended. For research purposes, the CS estimate from the 3 MT may be compared with the average value from the speed evoking gas exchange threshold (GET) and $VO_{2\text{max}}$ (50% Δ) during an incremental exercise test.¹² Others have examined VO_2 responses at constant speed bouts $<$ and $>$ the CS estimate; whereby, the bout $<$ CS should evoke a VO_2 steady state and the bout $>$ CS should evoke $VO_{2\text{max}}$ and culminate in fatigue (i.e., exercise in the severe domain).¹¹ Alternatively,

seeing the athletes perform their interval training bouts also can help in determining whether an athlete has paced. If an athlete had an inflated CS estimate, she/he will find it impossible to make the prescribed times for many of their intervals, and in particular, longer intervals. Conversely, if the 3 MT results are accurate, which they typically have been in my experience, the interval prescriptions will be very challenging but achievable.

All-out and Race Performances Conform to the Kinetic Energy Equation

The fastest running speeds for running different distances has been characterized as a “hyperbolic relationship.”³ Simply stated, the faster one’s running speed, the quicker the time to exhaustion, and an individual’s performance from the 3 MT can be modeled to construct both velocity-time and distance-time relationships (Figure 2). Interestingly, when first modeling the t_{LIM} s associated with different distances, I used the default unit of measurement for treadmills and GPS sensors in the USA (mph). Although the unit of measurement of $\text{mph} \cdot 150 \text{ s}$ is nebulous, when algebraically transformed, the calculation of t_{LIM} was still accurate. The units of measurements were indeed arbitrary because the relationship between running speed and t_{LIM} is proportionate and conforms to the kinetic energy equation.²

The kinetic energy equation is a useful construct for appreciating the true proportionality of CS and D' to total performance. To demonstrate conformity of the CS concept to the kinetic energy equation, consider the following example. If we project the t_{LIM} for Pre running at 1/2 of the difference between S_{150s} and CS, we would arrive at a speed of $6.67 \text{ m}\cdot\text{s}^{-1}$. The resulting t_{LIM} at that speed would double from 150 s to 300 s (i.e., 1/2 the difference equals double the t_{LIM}). A speed of $6.67 \text{ m}\cdot\text{s}^{-1}$ would cover an approximate distance of 2000 m, which is the product of $6.67 \text{ m}\cdot\text{s}^{-1}$ and 300 s. By doubling the time from 150 to 300 s, we derive that the energy from the CS

metric supported 1800 m (i.e., $6.0 \text{ m}\cdot\text{s}^{-1} \cdot 300 \text{ s}$) and D' supported 200 m (Figure 2, panel B). Equations 1 through 3 corroborate these same distances and speeds.

Using the same math described above, the 3 MT provides proportional energetic information, which may provide an understanding of training status. Using Pre's $S_{150 \text{ s}}$ performance as an example, the total projected distance he would have covered was 1,100 m (i.e., $7.33 \text{ m}\cdot\text{s}^{-1} \cdot 150 \text{ s}$) (note: this same datum point is illustrated in Figure 1, panel A). The fractional contribution of his D' would have been ~18% (i.e., his D' divided by the total distance covered, or 200 m/1,100 m). As distances are extended, the contribution of D' to total distance declines exponentially. Conversely, the contribution of CS to total performance increases exponentially, as illustrated with the 900 m distance bins in Figure 2, panel B. From the all-out test, it is evident that Pre had a high CS, with 82% of his energy for his $S_{150 \text{ s}}$ performance coming from his superior aerobic system. Indeed, a CS of $6 \text{ m}\cdot\text{s}^{-1}$ is shared by Gold Medal distance running winners within the past century.²⁰ As the CS concept is evaluated in more sports, we will gain a better idea of normative CS and D' data, and proportions that are optimal for different athletes and sports.

How the Critical Speed Concept May Inform Racing Strategy

When investigating the running 3 MT, two runners we observed had identical GET and $\text{VO}_{2\text{max}}$ values.¹² Using the metric of running economy (i.e., the percentage of GET relative to $\text{VO}_{2\text{max}}$),²¹ these runners may have been deemed “identical.” Yet one runner had a higher CS and a lower D' compared to the other (note: the athlete with the higher D' also had a higher speed evoking $\text{VO}_{2\text{max}}$). In their 3 MTs, the runner with the superior D' would cover more distance in the beginning of the test; however, the “identical” runner with the superior CS would ultimately

run a longer total distance over the span of 180 s. Such performance differences would indicate that each runner should utilize different racing strategies.

To explain racing strategy, the example of Pre using the 5,000 m event of the 1972 Olympic Games in Munich is useful. Leading up to the event, an epic race was anticipated by the up-and-coming collegiate runner Pre and the favored runner from Finland, Lasse Virén. As indicated, Pre had a CS of $6 \text{ m}\cdot\text{s}^{-1}$ and a D' of 200 m (Figure 1). Using the same procedure that formed Figure 1, Virén was modeled to have had a CS of $5.87 \text{ m}\cdot\text{s}^{-1}$ and a D' of 267 m. The notable disparities between these runners are as follows: Pre had a superior CS ($+0.12 \text{ m}\cdot\text{s}^{-1}$) whereas Virén has a superior D' ($+67 \text{ m}$). In layman's terms, Virén had a “kick” that was feared by his competitors. Indeed, his D' at that time exceeded 4 standard deviations above the mean D' of Gold Medal winning Olympic distance runners during that century.²⁰

With a D' of 200 m, Pre could have sustained a running speed $+0.25 \text{ m}\cdot\text{s}^{-1}$ relative to his CS of $6 \text{ m}\cdot\text{s}^{-1}$ for 5,000 m (See Figure 2, panel A). At a velocity of $6.25 \text{ m}\cdot\text{s}^{-1}$, covering a distance of 5,000 m, Pre's time would have been 13:20. That performance would have eclipsed the Gold winning performance by Lasse Virén of 13:23.4. With knowledge from the 3 MT, physiologists today would have told Pre that he was best advised to adopt a front running strategy, rather than holding back. A front running strategy involves running faster split-times within a given race rather than trying to “store energy” for a big finish. Mathematical modeling indicates that running below one's CS at any point during a middle distance event, will result in a slower time.¹⁰ By assuming a front running strategy, Pre would have forced Virén to rely on a large portion of his D' to keep pace. By utilizing a larger portion of his D' , Virén would have experienced a slow component of VO_2 ,²² which is associated with the recruitment of inefficient type II muscle fibers.⁵ Anecdotally, the front running athlete with a slightly superior CS can

strategically take away the “kick” of an opponent. Conversely, Virèn’s strategy in the 1972 race was perfect. By keeping the pack of runner’s closer to his pace, because they feared the pace he would dictate at the end of the race, Virèn took over the race in the last 200 m with his superior D' (note: a video of that race may be viewed online at YouTube by searching “Prefontaine 1972”). Knowledge of the CS concept was established since the late 1980’s and yet application to racing strategy is not pervasive in sports to date.

Prescription of High-Intensity Interval Training

Perhaps the most intriguing utility of the CS concept is the ability to prescribe and evaluate HIIT. Earlier, “identical runners,” in terms of their GET and VO_{2max} were mentioned, yet one had a superior D' and the other had a superior CS. Such differences beg the question: why would we prescribe these athletes the same HIIT training program when they have distinct deficiencies to develop? The most prevalent of methods for standardizing HIIT (e.g., %HR_{max}, %VO_{2max}) allow only for consideration of aerobic fitness. Prescribing HIIT using the CS concept involves the inclusion of D' within the prescription.

Programming HIIT with the CS concept can be simple. Using the 3 MT, the t_{LIM} associated with running a given speed or distance for each athlete can be predicted (Figure 2). For example, the t_{LIM} associated with running a distance of 1,000 m, using CS and D' data gathered from the 3 MT along with Eqn 3 can be estimated despite not having actually tested each athlete’s fastest time for running 1,000 m. Subsequently times for running 1,000 m intervals can be derived using modeled data. The equation for prescribing intervals for running overground (i.e., on a level track) is as follows:

$$\text{Interval } t_{LIM} = [D - (0.6 \cdot D')] / CS, \quad (\text{Eqn 5})$$

where D is distance, with our example being 1,000 m and 0.6 is an example fractional depletion of D' . By reducing D' within the equation, a longer time will be allotted to run the 1,000 m distance that is proportionate to each athlete's D' . Athletes with higher D' values can (and should) run at faster speeds relative to their CS because their superior D' enables them to do so. If their peer has a superior CS , the actual speed for that HIIT bout might be actually faster, but their interval speeds would be in closer proximity to their CS . Moreover, with standardized rest intervals, multiple bouts of these HIIT can be performed within a team practice setting (discussed below).

The prescription of HIIT using the CS concept relies upon the relationship between CS , D' , and the VO_2 slow component²². Specifically, at higher speeds relative to CS for any given athlete, D' expenditure will be more rapid along the gain in VO_2 toward VO_{2max} (i.e., a more rapid VO_2 slow component). Figure 3 depicts two methods for utilizing the CS concept for HIIT prescription. Panel A of Figure 3 depicts a prescription where 60% of D' is expended over a period of 300 s. Panel B depicts a prescription where the same 60% of D' is expended in the shortened period of 180 s (i.e., 60% of 300 s, in accordance with the kinetic energy equation). Thus, as the proportional depletion of D' between the two HIIT bouts is identical (i.e., the geometric area of the rectangles in Figure 3 are the same), the same metabolic response will be observed. Specifically, rate of rise in VO_2 will be more rapid in the 180 s bout, and a similar end-exercise VO_2 will be observed for the 300 s bout.²³ By modifying Eqn 3, the CS concept can be used to model interval speed-time performances (dashed curved lines in Figure 3). To derive specific speeds for a running specific interval training times, the following equation may be used:

$$\text{Interval } S = (D' \cdot 0.6)/t_{LIM} + CS. \quad (\text{Eqn 6})$$

One should note that the speeds for HIIT bouts on a treadmill should be adjusted when relying upon an overground running 3 MT. Specific considerations exist for speed and grade.²⁴ A previously published conversion table of outdoor speed in comparison to treadmill speed and grade is available,¹² and we have used this table successfully when prescribing treadmill HIIT bouts based on an overground running 3 MT.⁶

When implementing HIIT bouts for an entire team, the CS concept enables standardizing the rest intervals. For example, as shown in Figure 3, one group of athletes running a full 300 s (panel B) is focusing on improving CS without regard for D' . Conversely, a second group of athletes are running 180 s intervals (panel A) with the goal of improving CS with minimal decline in D' . As each group is depleting 60% of D' , the same time for the rest interval would be prescribed (e.g., 300 s); however, the work: rest ratio would vary. By enabling similar rest intervals, the rise in VO_2 toward $\text{VO}_{2\text{max}}$ for each interval will be similar between groups, despite different speed-time configurations.²³

To date, the HIIT prescriptions evaluated by our laboratory have been confined to 60, 70, and 80% D' depletion schemes, between 90 and 300 s. Intervals exceeding 300 s are too slow for training in the severe exercise domain (i.e., too slow to evoke $\text{VO}_{2\text{max}}$).^{12, 25} Conversely, intervals shorter than 90 s exceed the limit of the CS concept. That is, the speeds prescribed for HIIT bouts < 90 s may cause fatigue prior to the body's ability to expend D' and for VO_2 kinetics to reach $\text{VO}_{2\text{max}}$ due to muscular acidosis.²⁶

For 60% D' depletion, 4 to 5 intervals between 90 and 300 s are recommended (4 bouts when starting out), with a 1:1 work: rest ratio. For 80% D' depletion, 3 intervals between 90 and 300 s are recommended, with 1:1.5 work: rest ratio.⁶ Although we have used 70% D' depletion intervals, the sensitivity of the CS concept may exceed 10% (i.e., the metabolic responses

evoked by 70 and 80% expenditure do not differ).²³ Indeed, the test-retest reliability may be as high as 20%,²⁷ and although 70 and 80% intervals can be prescribed, mathematically, that difference in intensity is unlikely to evoke notably different metabolic responses. However, such observations were confined to cycle ergometry and may not extend to running. Moreover, subjects with higher D' values may be better able to comply with these HIIT prescriptions. Finally, HIIT bouts >80% D' depletion are not recommended, because given up to 20% variance in D' is possible, fatigue may occur very early on for bouts exceeding 80% D' depletion.²³

The prescription of 3 to 5-min HIIT bouts tend to favor gains in CS ⁶ and VO_{2max} ,²⁸ but do so at the expense of D' (i.e., D' declines). Bouts of 90 s yield smaller gains in CS with the preservation of D' ,⁶ and such bouts may appeal to athletes seeking to improve endurance without compromising sprinting capacity. A total of 4 weeks is sufficient to recognize training gains with previously trained athletes, and retesting of the 3 MT may be necessary in order to adjust HIIT prescriptions.

Cross-sectionally, D' may have a strength component, in that athletes engaging in anaerobic sports tend to have superior D' , or W' in cycling.¹ Indeed, two separate investigative groups have observed that a resistance exercise program improved their t_{LIM} for severe exercise; yet improvements were attributed to gains in W' with no change in CP .^{29, 30} The CS concept offers an opportunity to systematically explore the combined effects of aerobic and anaerobic training, on overall performance. More work is needed on the dose-response relationship of HIIT on CS parameters.

Some athletes may have trouble making their prescribed interval times with the above described methods. If running outdoors, aspects such as head winds and changes in gradient may reduce performance and should be considered when using the CS concept to prescribe

intervals. If athletes continue to experience difficulty complying with the HIIT prescriptions, it may be necessary to re-administer the 3 MT. A final consideration is diet, and in particular muscle glycogen levels. When athletes are in a glycogen-depleted state, the result will be a marked decline in W' with no effect on CP ,³¹ and presumably a decline in D' with no change in CS . Athletes should be instructed to eat in a similar manner to which they would before a competitive race. Completion of these HIIT prescriptions may not be achieved without proper attention to nutrition.

The greatest appeal of prescribing HIIT bouts using the CS concept is that the method provides speeds or times for intervals specific to the fitness level of each athlete. When a team of athletes are following these prescriptions, each athlete should experience a similar level of fatigue upon completion of each interval. As opposed to prescribing intervals as a percentage of maximum, the use of heart rate monitoring enables the sport scientist to evaluate the time-dependent increases in heart rate, that are characteristic of exercise intervals in the severe domain.⁶ The mainstream view of HIIT has focused on repetitive all-out regimes which involve declining intensities with each interval.³² Such HIIT regimes have come under scrutiny for adherence.³³ Conversely, times for intervals, or interval lap times, using the Eqn 5 and 6, provide quantitative goals that are challenging, yet attainable.⁶ By providing HIIT prescriptions with a quantitative goal, rooted in metrics of the CS concept, adherence for HIIT may be better for athletes and the general population alike.

Conclusions

The 3 MT provides an estimate of CS and D' on a group of athletes in a short period of time. Speed-time or distance-time profiles can be modeled using either previous PR data or by using the results of a running 3 MT. Information from the 3 MT identifies the mechanical

measure associated the maximal VO_2 steady-state (i.e., CS) and the capacity for running at speeds exceeding CS (i.e., D').⁴ Using Eqn 2, race times associated specific distances may be estimated. Competing runners may be compared using their CS and D' , and such comparisons may inform racing strategy (e.g., a runner with a superior CS should adopt a front running strategy).¹² . More research is needed on CS and D' normative values for a wider range of athletes, sports, and demographic information (e.g., competitive level, gender, playing position). Such information would inform HIIT training goals (e.g., what value of D' should a professional male rugby player be striving for?). Finally, the CS concept is a useful method for prescribing and evaluating the effects of HIIT. Prescribing HIIT using the CS concept allows for the consideration of an athlete's D' , a procedure that is a departure from methods of prescription based solely on percentages of maximal aerobic power. The CS concept offers a unique procedure for developing conditioning programs for middle distance runners and team-sport athletes; but may prove to have utility in other areas.

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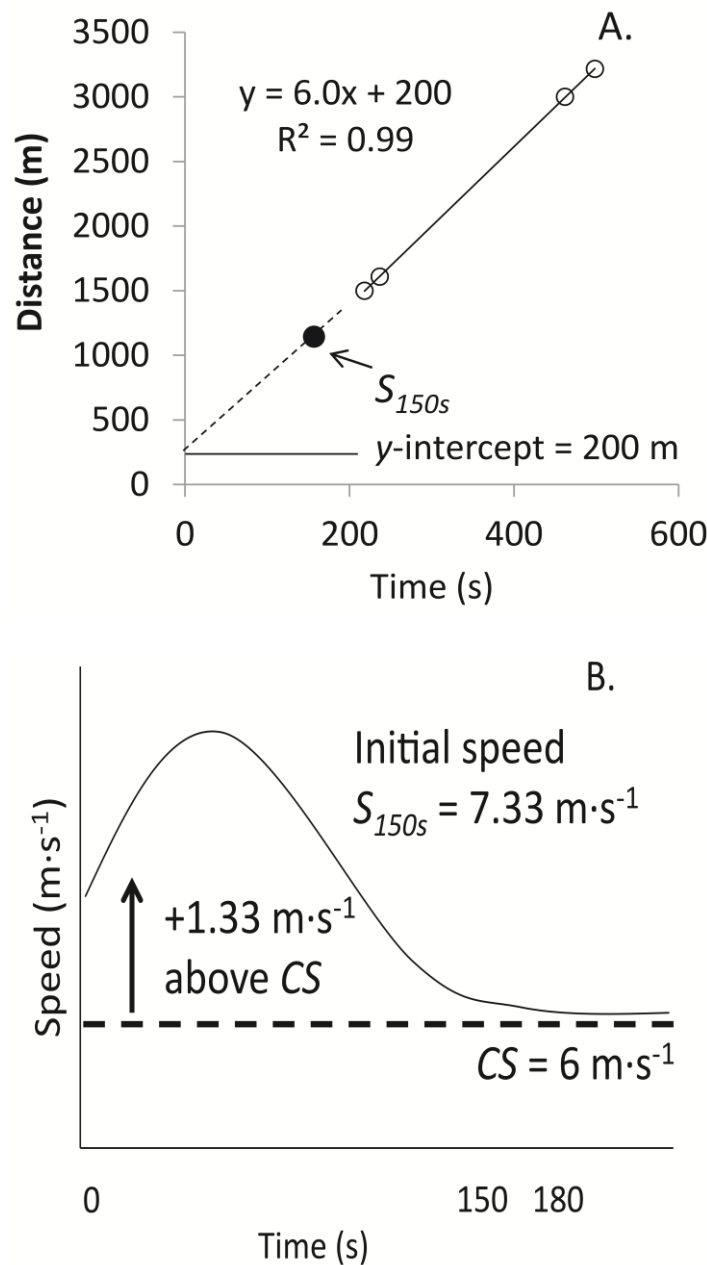
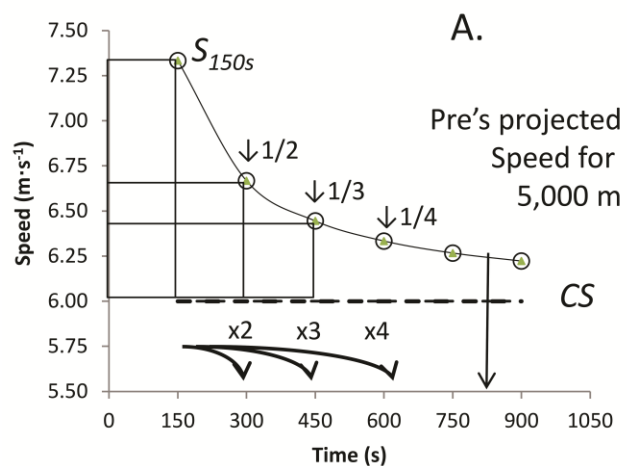


Figure 1. Linear regression of four personal records of Steve Prefontaine (Pre) (panel A). The slope reveals his critical speed was $6 \text{ m}\cdot\text{s}^{-1}$ and the y-intercept reveals his D' was 200 m. Panel B is used to illustrate how Pre would have performed in the 3-min all-out exercise test. The average speed in the last 30 s would be $6 \text{ m}\cdot\text{s}^{-1}$. His initial speed for 150 s (S_{150s}) would equal $7.33 \text{ m}\cdot\text{s}^{-1}$ and estimate of D' would equal 200 m using Eqn 4. The datum for S_{150s} also is plotted in panel A (i.e., a distance of 1,100 m covered in 150 s) to illustrate the conformity of all-out running performance in comparison to the four race performances.



$$6.00 \text{ m} \cdot \text{s}^{-1} \cdot 150 \text{ s} = 900 \text{ m}$$

B.

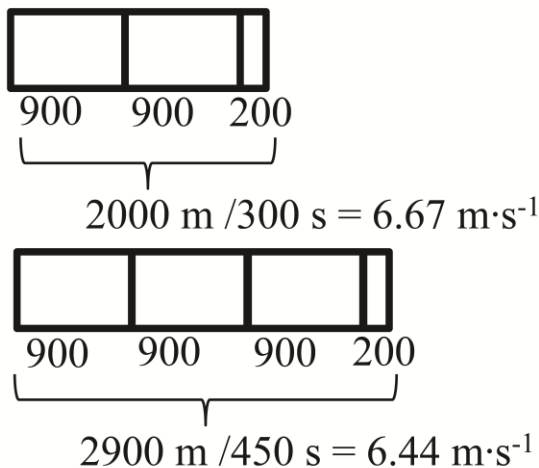


Figure 2. The modeling of individual speed-time and distance-time performances using Steve Prefontaine’s (Pre) data. Panel A depicts six separate data points of the curve which can be drawn using Eqn 2. The horizontal dashed line denotes critical speed (CS). Note that the geometric areas within the rectangles and square are equal, with the top right corner of each shape touching the data points for 150, 300, and 450 s. **As speed relative to CS is reduced proportionately, time is increased proportionately, as indicated by the x2, x3, and x4 labels.** The arrow intersects with an expected time of 800 s for “Pre” to run a distance of 5,000 m. Panel B illustrates the proportionality of the distance-time relationship. **The 900 m bins represent the running distance supported by CS. The 200 m bin represents the D’.** Take notice that as the total distance is divided by time, the speeds revealed are equivalent to the data points adjacent to the ↓1/2 and ↓1/3 labels.

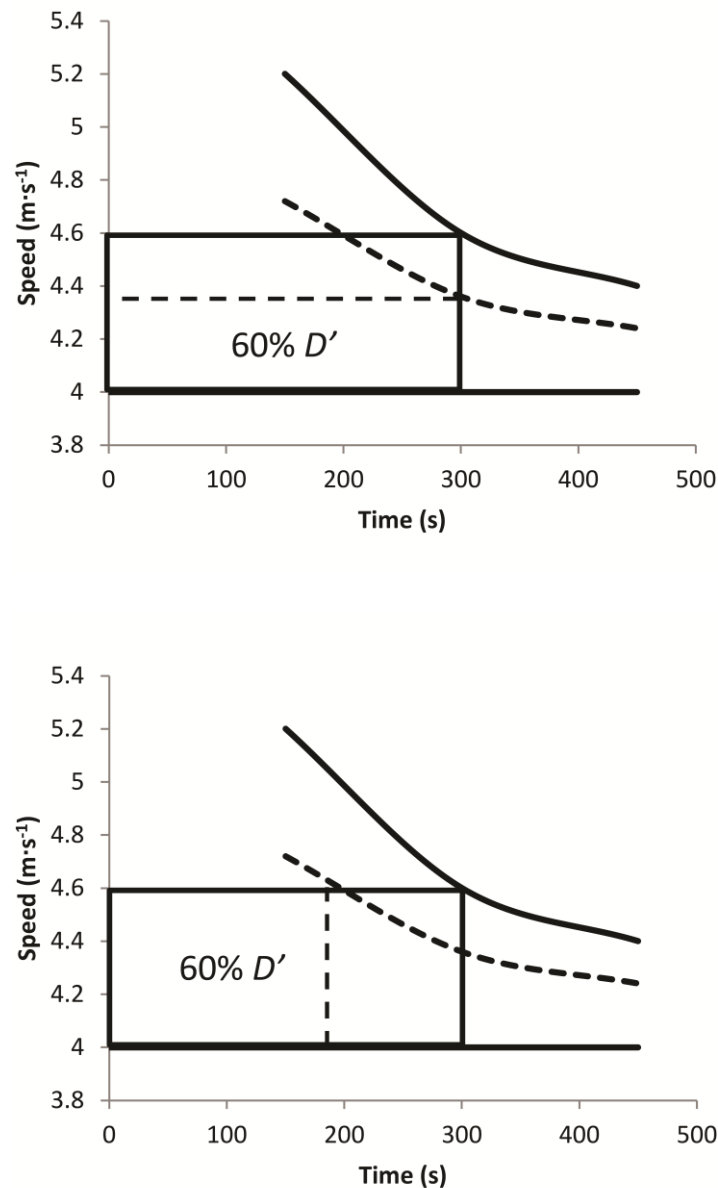


Figure 3. Speed-time curves (solid curves) for an athlete with a S_{150s} of $5.2 \text{ m}\cdot\text{s}^{-1}$, a CS of $4 \text{ m}\cdot\text{s}^{-1}$ and a D' of 180 m. The top corner of each solid rectangle is touching the speed associated with reaching a t_{LIM} of 300 s (i.e., $\frac{1}{2} S_{150s}$ and CS). The dashed curves were created using Eqn 6, where D' was multiplied by 0.6, causing a downward and leftward shift of the original speed-time curves. Panel A (dashed horizontal line) depicts a HIIT bout where speed is reduced 60%, relative to CS (i.e., a proportionately slower speed for the 300 s). Panel B (dashed vertical line) depicts a HIIT bout where speed is maintained but time is reduced proportionately by 60%. Each bout yields the same metabolic response